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Genetically Simulated GLP-1 Receptor Agonism and Cerebral Small Vessel Disease

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Abstract

Background and Objectives

Cerebral small vessel disease (cSVD) is a common cause of stroke and dementia without available definitive treatments. Glucagon-like peptide-1 receptor (GLP-1R) agonists have revolutionized the management of diabetes and obesity and have shown benefits in reducing cardiovascular risk, but it remains unknown if they could lower the burden of cSVD manifestations. Here, we investigated associations between genetic variants that mimic the action of GLP-1R agonists and cSVD phenotypes.

Methods

We applied a drug target Mendelian Randomization (MR) analysis using single-nucleotide polymorphisms (SNPs) within and near the *GLP1R* gene associated with reductions in HbA1c levels and body mass index (BMI). Our primary outcomes included the clinical end point of small vessel stroke (13,620 cases and 1,503,898 controls) and the imaging readout of white matter hyperintensity (WMH) volume (N = 48,454). The primary method used was inverse variance-weighted MR.

Results

Our *GLP1R* instruments consisted of 13 SNPs associated with HbA1c and 4 SNPs associated with BMI. After correcting for multiple comparisons, a genetically proxied reduction in HbA1c levels (per 6.75 mmol/mol) through perturbation in *GLP1R* was associated with lower odds of small vessel stroke (OR: 0.36, 95% CI 0.17–0.77, $p = 0.008$) and lower WMH volume on MRI (standardized beta: −0.52, 95% CI −0.84 to −0.19, $p = 0.002$). We observed similar, directionally consistent associations for genetically proxied *GLP1R*-driven BMI reduction (OR per 4.8 kg/m^2 decrement for small vessel stroke: 0.18, 95% CI 0.03–0.95, $p = 0.04$; standardized beta for WMH volume: −0.75, 95% CI −1.50 to 0.00, $p = 0.05$).

Discussion

Our study found that genetically proxied GLP-1 receptor agonism is associated with lower burden of clinical and imaging cSVD outcomes. These findings provide a rationale for clinical trials evaluating GLP-1 receptor agonists as a potential strategy to prevent cSVD progression.

Introduction

Cerebral small vessel disease (cSVD) refers to a group of pathologies affecting the small, perforating arterioles, capillaries, and venules of the brain, leading to a variety of clinical manifestations, such as ischemic and hemorrhagic stroke, cognitive impairment, gait dysfunction, urinary incontinence, and behavioral disorders.[1-6] cSVD pathologies are assessed through neuroimaging markers, such as white matter hyperintensities (WMH), enlarged perivascular spaces, lacunes, cerebral microbleeds, and small subcortical infarcts.[1] The burden of

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Glossary

BMI = body mass index; **cSVD** = cerebral small vessel disease; **GLP1R** = glucagon-like peptide-1 receptor; **GWAS** = genome-wide association study; **HbA1c** = glycated hemoglobin; **MR** = Mendelian Randomization; **RCT** = randomized controlled trial; **SNP** = single-nucleotide polymorphism; **WMH** = white matter hyperintensity.

cSVD increases substantially with age, with some MRI evidence of cSVD being present in almost all individuals older than 80 years.[7,8] Despite its public health importance, we lack definitive therapies that prevent cSVD progression.

Type 2 diabetes and obesity are established risk factors for the development and progression of cSVD.[9-12] Glucagon-like peptide-1 receptor (GLP-1R) agonists, or GLP-1RA, widely used to treat both conditions,[13] have been shown in randomized controlled trials (RCTs) to reduce cardiovascular risk,[14,15] including the risk of stroke.[16-18] Notably, GLP-1R agonists have demonstrated superiority over other antidiabetic drugs in preventing stroke.[19,20] Therefore, GLP-1R agonists may represent promising candidates for repurposing as therapeutic agents in cSVD.

Human genetics provide causal insights into mechanisms underlying predisposition to human disease, pointing to promising therapeutic targets[21] and informing drug development.[22] Genetically supported drug targets are 2–3 times more likely to lead to approved drugs.[23] Here, we leveraged large-scale genomic data sets to identify genetic variants in the *GLP1R* locus that mimic GLP-1 receptor agonism. We then conducted drug target Mendelian randomization (MR) analyses assessing the effects of these variants on cSVD phenotypes. Our aim was to investigate whether there is genetic support for GLP-1 receptor agonism as a promising strategy for preventing cSVD progression.

Methods

This study is conformant with the Strengthening the Reporting of Observational Studies in Epidemiology using Mendelian Randomization (STROBE-MR) statement[24] (eTable 1). An illustrative summary of the study design is presented in Figure 1.

Instrument Selection

To proxy the effects of GLP-1R agonists, we selected 2 sets of instruments comprised of genetic variants mimicking the well-established anti-diabetic and weight-lowering effects[13] of pharmacologic GLP-1R agonism. Valid instrumental variables must be robustly associated with the exposure (relevance assumption), remain independent of confounders (independence assumption), and influence the outcome only through the exposure (exclusion restriction assumption).[25] Using genome-wide association study (GWAS) summary statistics for glycated hemoglobin (HbA1c) levels (N = 389,889)[26] and body mass index (BMI, N = 806,834)[27] from individuals of European ancestry (eTable 2), we selected genetic variants based on the following criteria:

- Location within the *GLP1R* locus, defined as a window including the gene coding for GLP-1 receptor as well as 300 kb upstream or downstream to it (chromosome 6, base pair location: 39,016,574–39,055,519, based on the GRCh37-hg19 genome build)
- As no variants in the *GLP1R* locus were associated with BMI and only 2 with HbA1c at a genome-wide significance threshold ($p < 5*10^{-8}$), we used a less stringent threshold of $p < 5*10^{-5}$ to consider both antiobesity and antidiabetic effects of GLP-1R agonists.
- Lack of strong correlation due to linkage disequilibrium between the variants (clumped at $r^2 < 0.1$, clumping window = 10,000 kb).

For each individual variant, we calculated F-statistics[25] and the proportion of variance it explained in HbA1c or BMI[28] (eTable 3).

Instrument Validation

To validate the instruments and quantify their effects on HbA1c levels and BMI, we analyzed individual-level data from the UK Biobank. UK Biobank is a UK-wide prospective cohort of 503,317 adults aged 39–73 years, recruited between 2006 and 2010.[29] The baseline assessment of the UK Biobank occurred shortly after the first GLP-1R agonist for diabetes was approved and long before these drugs became widely used as antidiabetic or antiobesity medications. In addition, according to baseline medication records (UK Biobank data field "20003"), none of the participants was taking a GLP-1R agonist at the time of the initial assessment.

Among 487,067 participants with available genotypic data,[30] HbA1c levels were available for 462,736 and BMI for 485,095. Using the clumping and thresholding method, we calculated genetic risk scores for both instruments and explored differences in median HbA1c and BMI between the top and bottom percentiles of the distribution.[31] The confidence intervals were derived using bootstrapping with 10,000 replications. The scores were calculated with *plink v2.00a*.[32]

Furthermore, given the proven benefits of GLP-1R agonists against atherosclerotic cardiovascular disease and diabetes,[13-15] as positive controls, we assessed the effects of our instruments on coronary artery disease (CAD, 210,842 cases, 1,167,328 controls, ~85% of European ancestry)[33] and type 2 diabetes (T2D, 428,452 cases, 2,107,149 controls, 71.4% of European ancestry).[34] For this analysis, we used publicly available GWAS summary statistics, as presented in eTable 2.

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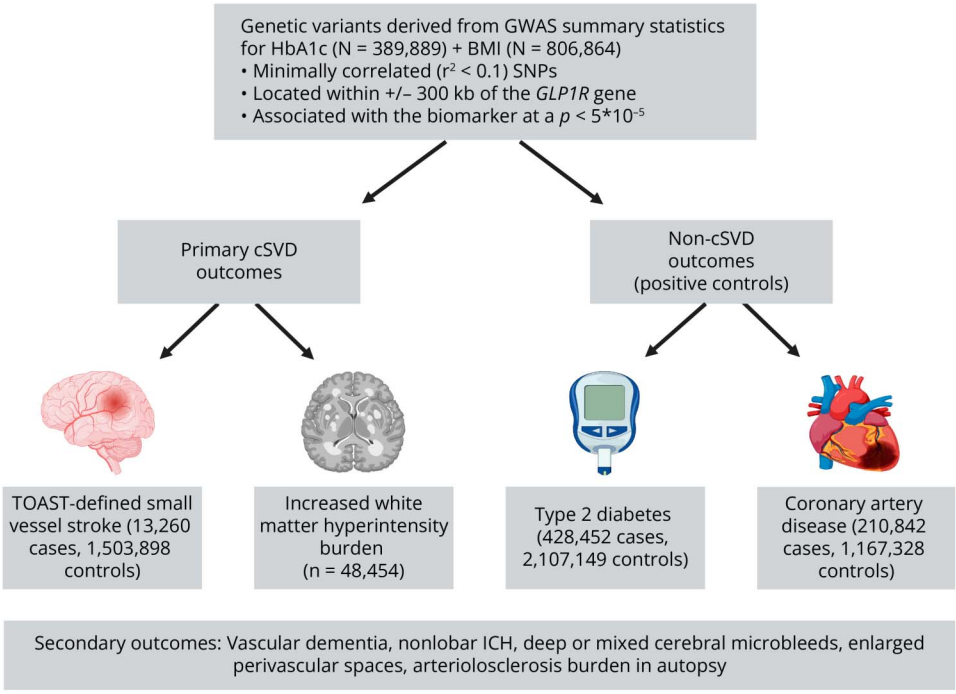
Figure 1 Graphical Overview of Study Design and Sample Sizes of the GWAS Used for the Analysis



BMI = body mass index; cSVD = cerebral small vessel disease; GLP1R = glucagon-like peptide-1 receptor; GWAS = genome-wide association study; HbA1c = glycated hemoglobin; ICH = intracerebral hemorrhage; SNPs = single nucleotide polymorphisms; TOAST = Trial of ORG 10172 in Acute Stroke Treatment. Created in BioRender.com.

Outcomes

Our primary outcomes included the clinical end point of small vessel stroke (SVS, or lacunar stroke), as defined by the TOAST criteria,[35] and the neuroimaging readout of WMH volume on MRI (N = 48,454). The 2 outcomes represent the most common manifestations of cSVD[36,37] and have been studied as end points in clinical trials.[2] We derived the effects of the variants comprising our genetic instruments on these outcomes from publicly available GWAS summary statistics. For SVS, we used data from the GIGASTROKE study (13,620 cases and 1,503,898 controls, 82% of European ancestry)[38] and for WMH volume from a meta-analysis of 48,454 individuals (95% of European ancestry) from the UK Biobank and 23 CHARGE cohorts.[39]

As secondary exploratory outcomes we tested associations with vascular dementia (3,892 cases, 466,606 controls),[40] nonlobar intracerebral hemorrhage (881 cases, 1,481 controls),[41] mixed or strictly deep cerebral microbleeds (1,293 cases, 24,569 controls),[42] enlarged perivascular spaces (9,189 cases, 30,811 controls),[43] and arteriolosclerosis burden in autopsied study participants (N = 6,668).[44] Data sources for each outcome are presented in eTable 2.

Statistical Analysis

We performed two-sample MR analyses using the *TwoSampleMR* R package (version 0.6.15). We used fixed-effects inverse variance weighted MR as our principal analytical approach.[45] The Cochran Q test was used to assess presence of between-variant heterogeneity in the derived effect estimates with $p < 0.05$ indicating significant heterogeneity.[46] The weighted median estimator and MR Egger regression were used as sensitivity analyses to test robustness against the main MR assumptions. The weighted median estimator provides a consistent estimate of the causal effect, as long as at least 50% of the instrumental variables are valid.[47] MR Egger regression assumes that the effect of each instrument with the exposure is independent of any pleiotropic effect.[48] An Egger intercept significantly different from zero ($p < 0.05$) provides an indication of horizontal pleiotropy.

The derived odds ratios and beta coefficients correspond to a 6.75 mmol/mol reduction in HbA1c levels and 4.8 kg/m^2 reduction in BMI. The statistical significance threshold was set at $p < 0.025$ correcting for multiple comparisons for our 2 outcomes with the Bonferroni approach. A $p < 0.05$ indicated nominal significance.

Standard Protocol Approvals, Registrations, and Patient Consents

The UK Biobank has obtained approval from the Northwest Multi-Center Research Ethics Committee (REC reference for UK Biobank is 11/NW/0382). All participants have provided written informed consent. The publicly available GWAS summary statistics were generated by studies that received ethical approval and participant consent for both the analysis and distribution of summary-level data, as detailed in the original publications.

Data Availability

The GWAS summary statistics for the exposures and outcomes used in this study are described and cited above, with

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further details presented in eTable 2. No new data were generated. UK Biobank data are available on application to the UK Biobank.

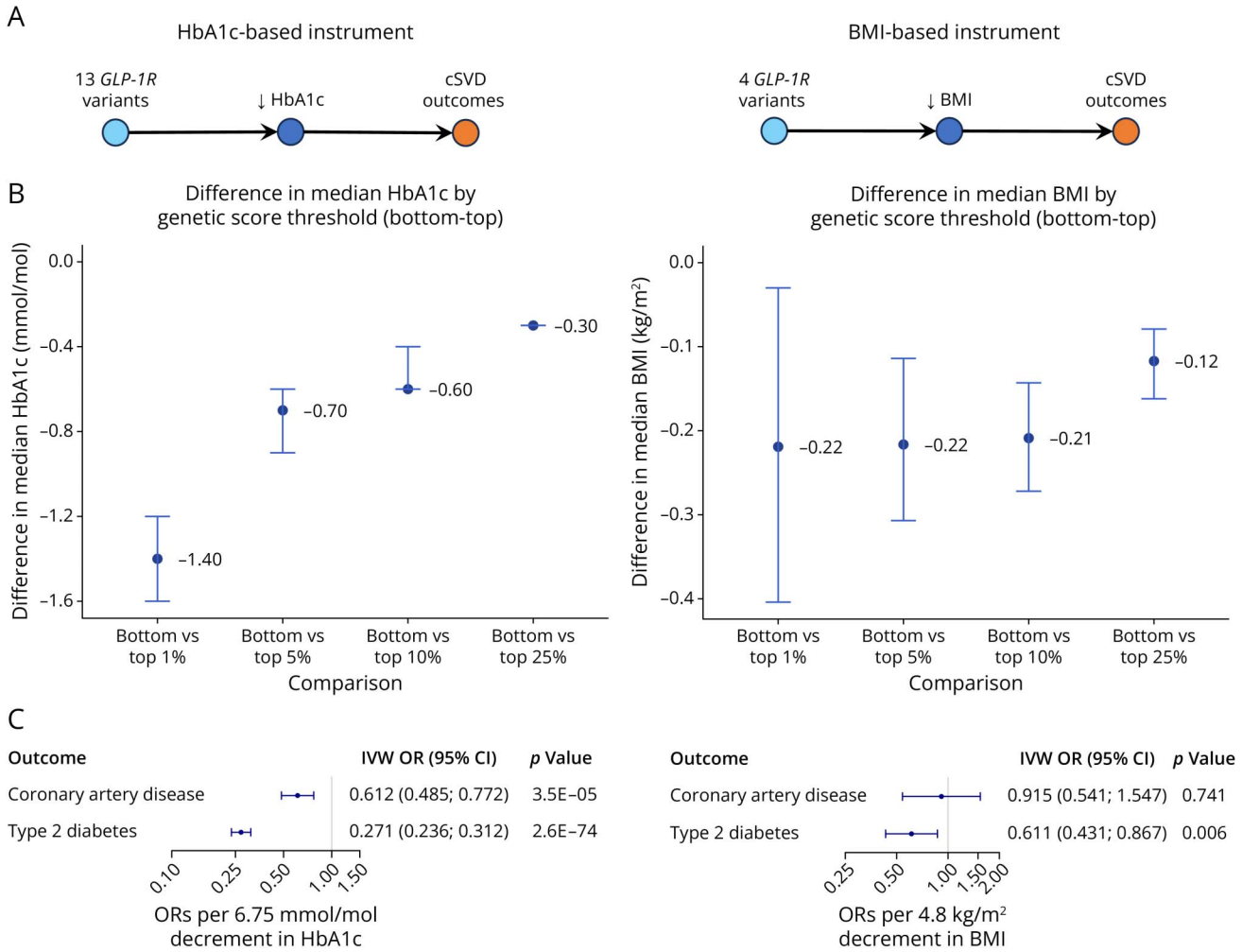
Results

We detected 13 and 4 genetic variants in the *GLP1R* locus meeting our selection criteria for the HbA1c and BMI instrument, respectively (Figure 2A). The F-statistics of all variants were >10, a threshold commonly used to indicate sufficient instrument strength[25] (eTable 3). The 2 genetic instruments explained 0.09% and 0.01% of the variance in HbA1c and BMI, respectively (eTable 3). In the UK Biobank, genetic scores built from these instruments showed significant differences between the extremes of the score distributions. Compared with the bottom 1%, individuals in the top 1% had a 1.4 mmol/mol lower median HbA1c (95% CI −1.6 to −1.2) and a 0.22 kg/m^2 lower median BMI (95% CI −0.40 to −0.03). (eTable 4, Figure 2B).

In line with clinical trials,[13-15] we found genetically proxied HbA1c-lowering through *GLP1R* perturbation to be associated with lower odds of T2D (OR per 6.75 mmol/mol decrease in HbA1c: 0.27, 95% CI 0.24–0.31, $p = 2.6*10^{-74}$) and lower odds of coronary artery disease (OR: 0.61, 95% CI 0.49–0.77, $p = 3.5*10^{-5}$). The BMI-instrument was also significantly associated with T2D (OR per 4.8 kg/m^2 in BMI: 0.61, 95% CI 0.43–0.87, $p = 0.006$), but not with CAD (OR: 0.92, 95% CI 0.54–1.55, $p = 0.741$) (Figure 2C, eTables 5 and 6).

Genetically proxied GLP-1R agonism as instrumented by HbA1c-lowering variants was significantly associated with both lower odds of TOAST-defined SVS (OR per 6.75 mmol/mol decrease in HbA1c: 0.36, 95% CI 0.17–0.77, $p = 0.008$)

Figure 2 HbA1c- and BMI-Based Genetic Instruments of GLP-1 Receptor Agonism Used in Our Analyses



(A) Theoretical framework of our analyses. (B) Results from the individual-level data analysis in the UK Biobank, with differences in median HbA1c/BMI between the top and bottom 1%, 5%, 10% and 25% of the distribution of genetic risk scores derived from the selected instruments. The 95% confidence intervals were derived using bootstrapping. (C) Inverse-variance weighted Mendelian Randomization of the effects of genetically proxied GLP-1 receptor agonism on positive outcomes of coronary artery disease and type 2 diabetes. BMI = body mass index; cSVD = cerebral small vessel disease; GLP1R = glucagon-like peptide-1 receptor; HbA1c = glycated hemoglobin; IVW = inverse-variance weighted; OR = odds ratio.

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and lower WMH volume on MRI (standardized beta per 6.75 mmol/mol decrease in HbA1c: −0.52, 95% CI −0.84 to −0.19, p = 0.002) (Figure 3, A and B, eTables 5 and 6). There was no evidence of between-variant heterogeneity (all p-values>0.05) and the results were largely consistent in sensitivity analyses. Specifically, the weighted median MR effects were significant (standardized beta: −0.64, 95% CI −1.06 to −0.21, p = 0.003 for WMH volume, OR: 0.30, 95% CI 0.11 to 0.83, p = 0.020 for SVS), whereas the effect estimates for MR Egger were highly consistent, albeit nonsignificant (standardized beta: −0.60, 95% CI −1.37 to 0.17, p = 0.163 for WMH volume, OR: 0.31, 95% CI 0.05 to 1.97, p = 0.251 for SVS) without evidence of significant intercepts that would indicate risk of directional pleiotropy (Figure 3, C and D, eTables 5 and 6). The effects of genetically proxied GLP-1 receptor agonism on SVS and WMH volume were also consistent when using the instrument consisting of the 4 BMI-lowering variants (Figure 3, A and B, eTables 5 and 6).

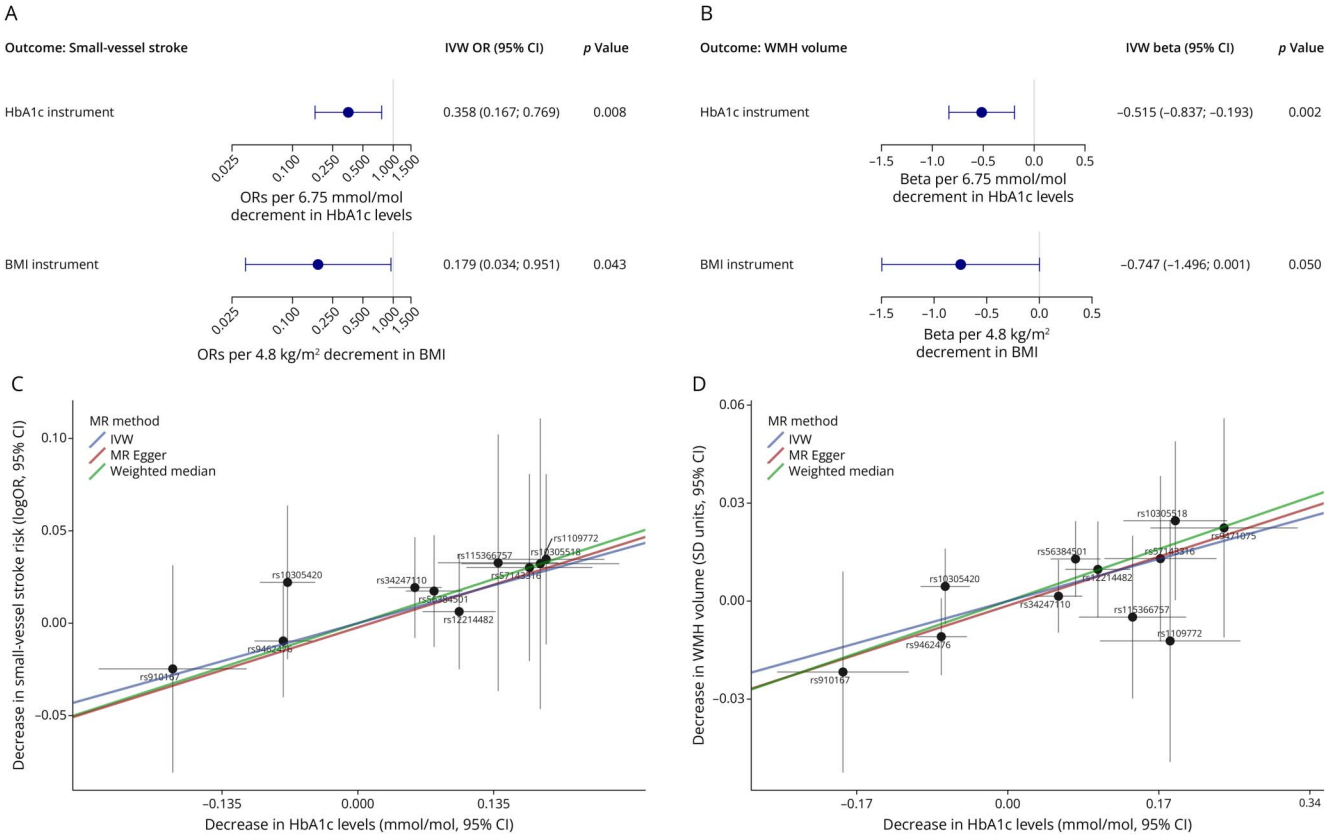
We found no evidence of significant associations with the secondary exploratory clinical, imaging, or pathologic cSVD outcomes (eTables 5 and 6). This may be related to limited statistical power, as indicated by the very wide 95% confidence intervals of the derived effect estimates, which included largely implausible values.

Discussion

Our drug target Mendelian randomization study supports that genetic variants mimicking GLP-1 receptor agonism are associated with lower lifetime burden of clinical and imaging manifestations of cSVD. Previously published RCT meta-analyses[16-18] demonstrated a GLP-1RA-mediated decrease in ischemic stroke risk but have not studied their effects on cSVD manifestations. At the moment, there is only one registered ongoing phase 2 clinical trial investigating the probable role of GLP-1RA in preventing the progression of cSVD (GAPP-SVD, NCT05356104). Considering the lack of disease-modifying cSVD therapeutic regimens, our genetic data support the inclusion of cSVD-related manifestations as outcomes in GLP-1RA clinical trials.

Our study has specific limitations. First, although we obtained the largest available cSVD outcome GWAS data sets for sample size, we encountered a low statistical power issue with respect to secondary outcomes, restricting the ability of

Figure 3 Mendelian Randomization Results for Primary cSVD Outcomes



(A–B) Forest plots showing the effects of the HbA1c and BMI instruments on small-vessel stroke and WMH volume. (C–D) Scatterplots demonstrating the per-SNP effects of the HbA1c instrument on small-vessel stroke and WMH volume with regression lines for IVW, MR Egger and weighted median methods. BMI = body mass index; CI = confidence intervals; HbA1c = glycated hemoglobin; IVW = inverse-variance weighted; MR = Mendelian Randomization; OR = odds ratio; WMH = white matter hyperintensities.

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detecting significant MR estimates and assessing potential pleiotropy presence, as the Egger intercepts were accompanied by large deviations. This is likely attributable to the fact that the genetic instrument did not explain a sufficiently high proportion of the exposure variance and that the sample size of the underpowered outcomes was relatively smaller. Second, MR evaluates the lifelong consequences of GLP-1R activation, which may not necessarily correspond to the short-term pharmacologic effects of GLP-1RA. Third, despite the fact that we used minimally correlated variants within the *GLP1R* locus, we cannot completely rule out the presence of impacting neighboring genes or linkage disequilibrium. Fourth, the variant selection threshold was less stringent than the genome-wide significance threshold. This selection could introduce weak instrument bias, as the effects of the instruments on HbA1c levels and BMI in the UK Biobank are statistically significant but modest. Weak instrument bias would shift the estimates towards the null.[49] Nevertheless, we observed the expected effects of genetically proxied HbA1c and BMI reduction on clinical phenotypes such as T2D and CAD. Fifth, our genetic instruments did not include rare variants (minor allele frequency <0.01), which could have a meaningful impact on GLP1R-mediated reduction in HbA1c and BMI levels. Sixth, the vast majority of the study population was from European ancestry, potentially restricting the generalizability of our results to other ancestry groups.

In conclusion, human genomic data support the hypothesis that GLP-1 receptor agonism may reduce the burden of cSVD and lower risk of its complications. These findings support the conduct of RCTs to test the repurposing potential of GLP-1 receptor agonists as disease-modifying therapies for cSVD progression.

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Author Contributions

P. Zangas: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M. Omarov: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M.K. Georgakis: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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Disclosure

M.K. Georgakis reports consulting fees from Tourmaline bio, Inc., Pheiron GmbH, Dexcel Pharma Technologies Ltd. and GLG, Inc., all unrelated to this work. The other authors declare no competing interests. Go to Neurology.org/NG for full disclosures.

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